DAXDA Manual 2: Recursive Engine Specification

Operational architecture for claim challenge, assumption mapping, uncertainty, and loop control.

DAXDA - Dialectical Analysis x Dialectical Analysis

Positioning rule: sell the risk reduction, decision quality, transparency, and audit record. Keep recursion as the engine, not the headline.

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1. Engine Overview

The DAXDA engine should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A AI governance team buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind technical architecture is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is black-box trust without inspection. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. a structured pass from claim to audit log should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

- Define the input and state what decision or claim is being reviewed for engine overview.
- Separate mechanism language from buyer-facing outcome language.
- Record assumptions, counterarguments, missing evidence, and confidence status.
- Use a stop rule so review improves the decision instead of becoming theater.
- Produce an artifact that an outsider can inspect without needing the founder in the room.

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2. Claim Decomposition

Claim decomposition should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction,

assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A analyst buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind semantic parsing is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is treating a paragraph as one indivisible assertion. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. breaking outputs into testable claims should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for claim decomposition.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

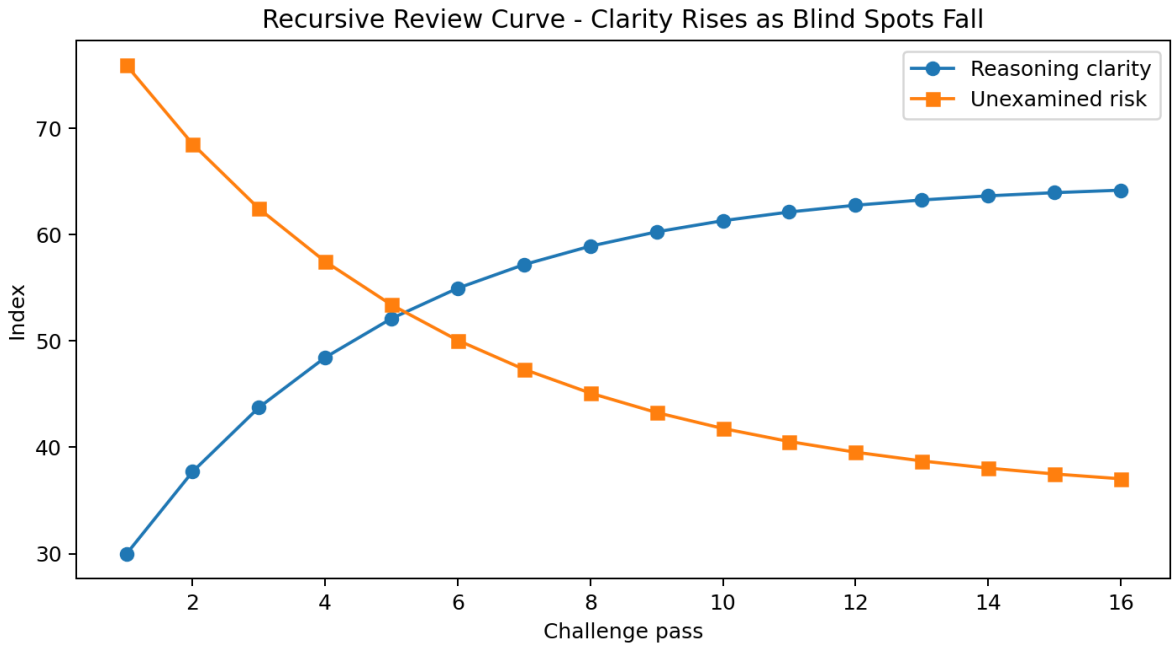


Figure: loop model for DAXDA deployment.

3. Assumption Mapping

Assumption mapping should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A research team buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind dependency inspection is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is letting hidden premises drive conclusions. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. marking each assumption as factual, normative, causal, or operational should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for assumption mapping.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

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4. Evidence Challenge

Evidence challenge should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A evaluator buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind verification discipline is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is evidence theater. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. linking each claim to source strength and contradiction status should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for evidence challenge.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

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5. Counterargument Generation

Counterargument generation should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A decision committee buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind devil advocate structure is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is groupthink with a professional haircut. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. forcing strongest opposing explanations into the record should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for counterargument generation.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

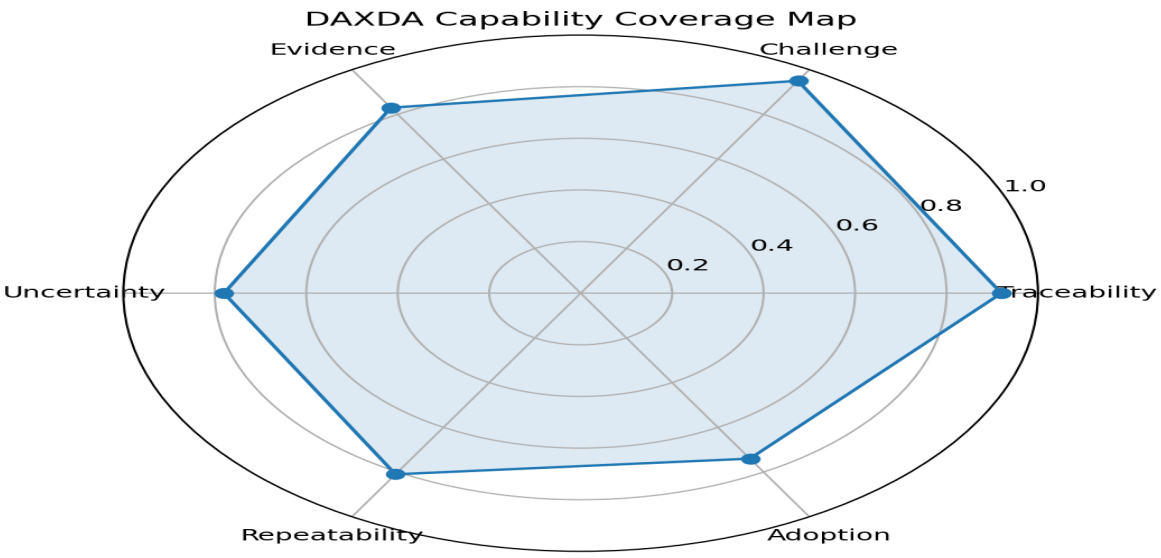


Figure: radar model for DAXDA deployment.

6. Uncertainty Calibration

Uncertainty calibration should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A risk manager buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind confidence governance is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is binary certainty claims. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not

merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. confidence bands, missing evidence flags, and reversible conclusions should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for uncertainty calibration.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

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7. Recursive Loop Control

Recursive loop control should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A operator buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind recursion management is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is over-recursion and analysis paralysis. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. stop rules based on marginal improvement and risk class should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a

way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for recursive loop control.

• Separate mechanism language from buyer-facing outcome language.

• Record assumptions, counterarguments, missing evidence, and confidence status.

• Use a stop rule so review improves the decision instead of becoming theater.

• Produce an artifact that an outsider can inspect without needing the founder in the room.

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8. System Boundary Conditions

System boundaries should be understood as a product discipline before it is understood as a theory. The practical buyer does not wake up asking for a recursive dialectical engine. The buyer has a narrower and more painful question: can this decision, claim, model output, policy recommendation, or research conclusion survive structured challenge? DAXDA answers that question by forcing the work to pass through repeated rounds of claim extraction, assumption inspection, evidence comparison, counterargument generation, uncertainty labeling, and final decision recording. The important commercial point is blunt: the mechanism is interesting, but the purchase motive is control. A technical buyer buys DAXDA because unmanaged reasoning risk has become expensive, reputationally dangerous, and operationally hard to document.

The strongest positioning is therefore outcome-first. DAXDA reduces unexamined risk, improves decision quality, creates a documented review trail, and makes disagreement usable instead of chaotic. The framework does not promise omniscience. That would be amateur hour. It promises a disciplined process for making claims more inspectable. In high-stakes environments, that distinction matters. A perfect answer is rarely available. A better challenged answer, with visible assumptions and a traceable audit path, is often exactly what the organization needs.

The operating principle behind product constraints is that every conclusion carries load-bearing assumptions. Some assumptions are factual, some are methodological, some are political, some are economic, and some are simply inherited habits. When those assumptions remain hidden, the organization confuses fluency with reliability. DAXDA interrupts that pattern. It asks what the answer depends on, what would break the answer, what evidence is missing, what alternative explanation could fit the same facts, and what decision record should exist after review. This is not decorative skepticism. It is a controlled risk process.

The main failure DAXDA is designed to counter is claiming more than the system can prove. That failure appears in different costumes: an AI model gives a confident answer that has not been checked; a leadership team agrees too quickly because dissent is socially costly; a research group cites a claim that depends on weak evidence; a procurement team selects a vendor without testing the counter-case. DAXDA turns these situations into structured review events. It does not merely say, 'think critically.' It specifies what to inspect, what to challenge, what to log, and how to decide whether the conclusion is ready to move forward.

The practice layer is where the product becomes real. explicit limitation statements and human signoff gates should be implemented as a repeatable workflow with named inputs, named outputs, decision thresholds, and an audit trail. Without that, DAXDA stays trapped as a clever concept. With it, DAXDA becomes a governance product: a way to prove that important thinking was challenged before it was trusted.

Operating checklist

• Define the input and state what decision or claim is being reviewed for system boundary conditions.

• Separate mechanism language from buyer-facing outcome language.

- Record assumptions, counterarguments, missing evidence, and confidence status.
- Use a stop rule so review improves the decision instead of becoming theater.
- Produce an artifact that an outsider can inspect without needing the founder in the room.